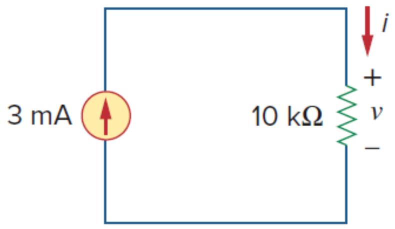
Chapter 2, Problem 2PP

Problem

For the circuit shown in Fig. 2.9, calculate the voltage *v*, the conductance *G*, and the power *p*.

Figure 2.9:



Step-by-step solution

Step 1 of 4

The circuit is shown in Figure 1.

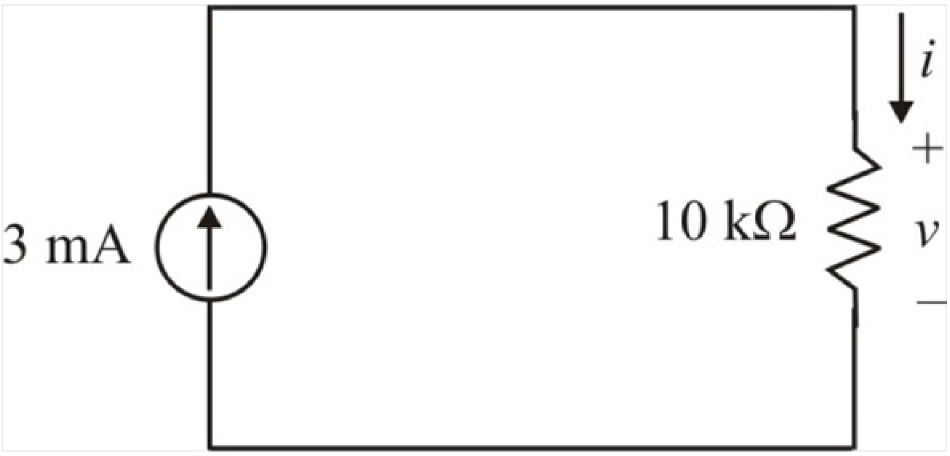


Figure 1

Step 2 of 4

The current flowing through the resistor is the same as the source current (3mA) because the resistor and the current source are connected in series. Hence the voltage across the resistor is:

Write Ohm's law.

$$\begin{aligned} v &= iR \\ &= \left(3\times10^{-3}\right)\left(10\times10^{3}\right) \\ &= 30\,\text{V} \end{aligned}$$

Therefore, the voltage is $\boxed{30\text{ V}}$.

Step 3 of 4

The reciprocal of the resistance is called conductance.

Calculate the conductance.

$$\begin{aligned} G &= \frac{1}{R} \\ &= \frac{1}{10\,\text{k}\Omega} \\ &= 100\,\mu\text{S} \end{aligned}$$

Therefore, the conductance is $\boxed{100\ \mu\text{S}}$.

Step 4 of 4

Calculate the power.

$$\begin{aligned} p &= vi \\ &= (30\,\text{V})(3\,\text{mA}) \\ &= 90\,\text{mW} \end{aligned}$$

Therefore, the power is $\boxed{90\ \text{mW}}$.